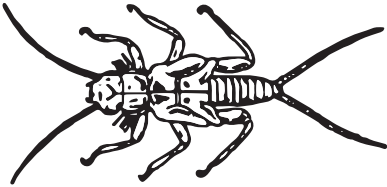
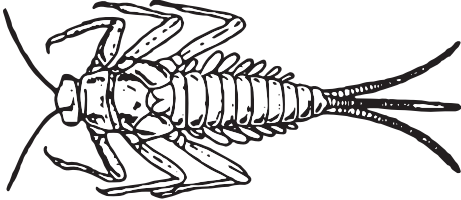
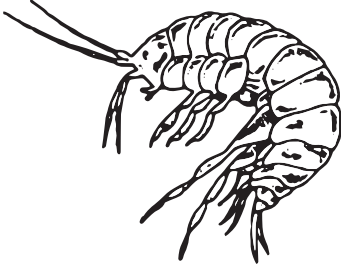
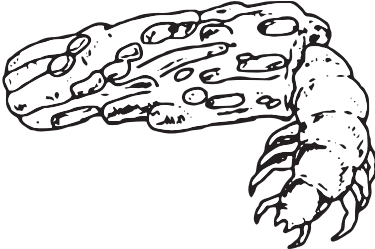
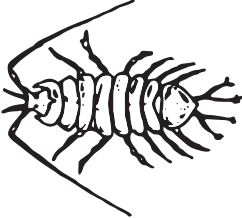
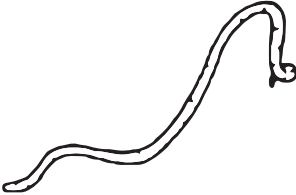
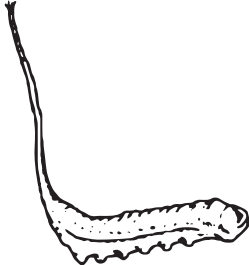


Examiner
only

3. Water pollution can be monitored using invertebrates as indicators. Some indicator species are shown below.

Water quality	Indicator species	
Clean water	 Stonefly nymph	 Mayfly nymph
Some pollution	 Freshwater shrimp	 Caddisfly larva
Moderate pollution	 Water louse	 Bloodworm
High pollution	 Sludgeworm	 Rat-tailed maggot
Very high pollution	No life	

(Not drawn to scale.)

(a) Describe how you would carry out an investigation to determine water quality in a stream using invertebrates as indicator species. [6 QER]

Examiner
only

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

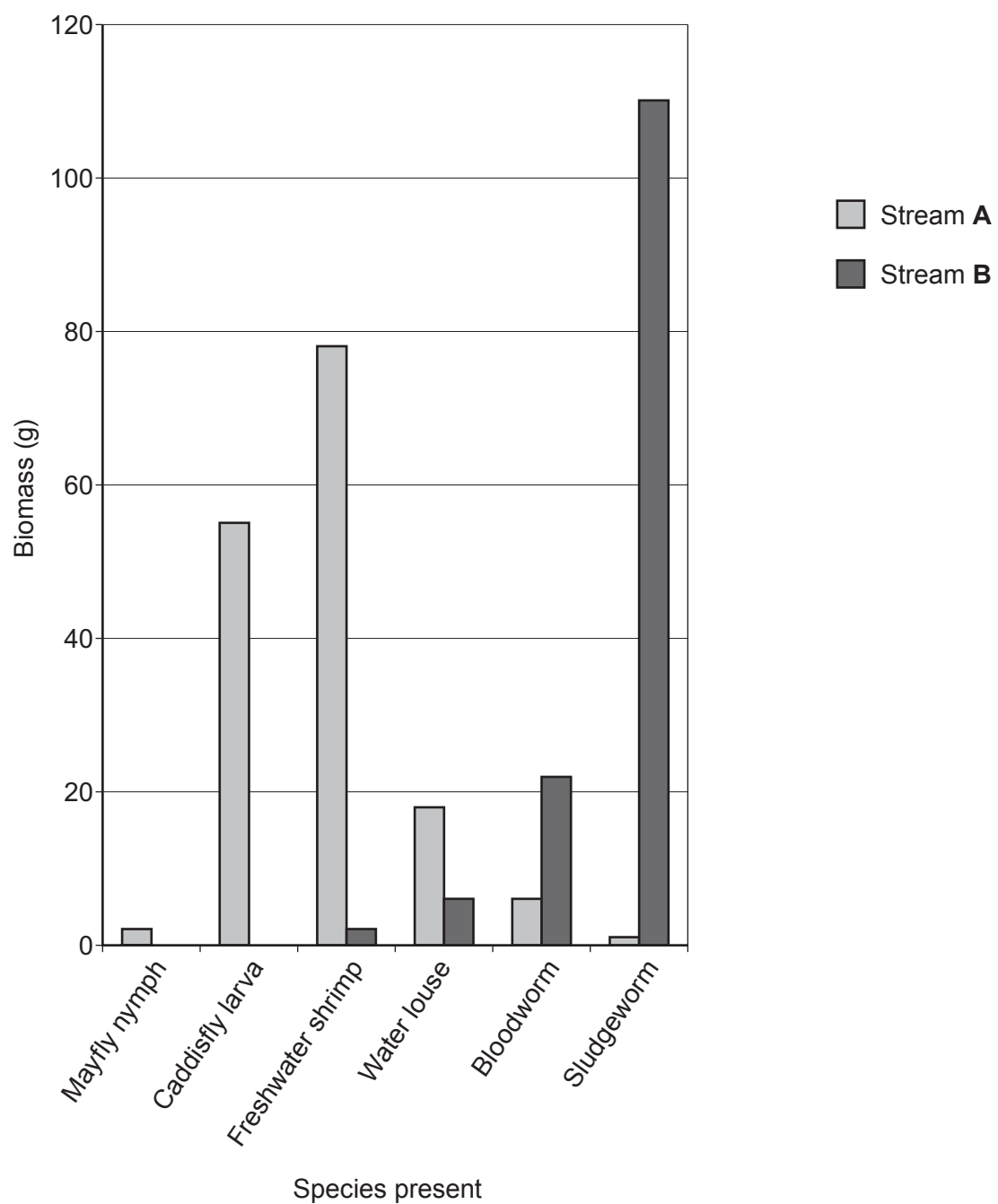
.....

.....

.....

.....

(b) The results from such an investigation are shown in the chart below.



- (i) Explain what conclusions can be made about the water quality in stream **A** and stream **B**. [4]

.....

.....

.....

.....

.....

.....

- (ii) Explain why it is difficult to obtain repeatable data in this experiment. [2]

.....

.....

.....

Examiner
only